

CHRISTINA SINGH

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SUMMARY

I am a second-year Ph.D. student at the University of Arizona's Lunar and Planetary Laboratory studying the distribution of boulders on mid-latitude Martian ice deposits to reconstruct Mars' paleoclimate history and constrain its water ice inventory. My broader interests include remote sensing, geomorphology, glacial and periglacial processes, and the application of machine learning to planetary surface analysis.

EDUCATION

Ph.D. Lunar and Planetary Laboratory, The University of Arizona, Expected 2030

Ph.D. in Planetary Science

Minor(s): Remote Sensing, Geomorphology

Advisor: Dr. Shane Byrne

B.S. Department of Physics, Astronomy, and Geophysics, Connecticut College, 2023

Major: Physics (concentration: Astrophysics); Minor(s): Mathematics, Geoscience

RESEARCH SKILLS

Computer Software: MaxIm DL, Microsoft Office (Word, Excel, PowerPoint), LaTeX, QGIS, ArcGIS (Desktop and Pro), JMARS, Google Earth Pro, SOCET SET, Ames Stereo Pipeline (ASP), Agisoft Metashape, Integrated Software for Imagers and Spectrometers (ISIS), Geospatial Data Abstraction Library (GDAL)

Dataset Experience: THEMIS, CTX, CRISM, HiRISE, MOLA, LROC, SHARAD

Computer Languages: Python, MATLAB

Computer Operating Systems: Windows 11, MacOS, Linux

RESEARCH EXPERIENCE

Lunar and Planetary Laboratory (LPL), University of Arizona

Description: Doctoral research project developing ROSCO (ROCK Shadow Counter), a semi-automated boulder detection pipeline, to map boulder distributions on Martian ice deposits and reconstruct Mars' paleoclimate history.

Tucson, AZ

September 2023

– Present

- Develops ROSCO's image processing and shadow analysis algorithms for automated boulder detection in HiRISE imagery
- Maps boulder distributions on viscous flow features (VFFs) to assess ice mobilization history and reconstruct Mars' paleoclimate
- Analyzes boulder abundances inside sublimation pits and on surrounding plains to constrain the sediment content and purity of Martian ice deposits

Advisor: Dr. Shane Byrne

Lunar and Planetary Institute (LPI) Summer Undergraduate Program for Planetary Research (SUPPR), NASA's Jet Propulsion Laboratory (JPL)

Description: Planetary geology research to characterize potential landing sites for future missions to retrieve sample cores acquired by the Perseverance Rover.

Pasadena, CA

June – August

2023

- Characterized the surface terrain of a potential sample return landing site
- Mapped rock size frequency distributions using ArcGIS Pro
- Developed MATLAB code to determine cumulative fractional areas

Advisor: Dr. Matthew Golombek

Co-advisor: Dr. Nathan Williams

Georgia Tech Broadening Participation in Atmospheric Science, Oceanography, and Geosciences Research Experience for Undergraduates (REU), Georgia Institute of Technology

Atlanta, GA
May - July 2022

Description: Planetary geology research with an emphasis on remote sensing, Mars geomorphology, stratigraphy, sedimentology, and paleoenvironments.

- Investigated the geomorphology of a distal remnant of the Hypanis delta
- Characterized stratigraphic layers of this deposit in QGIS
- Developed Python code to compute orientations of layers

Advisor: Dr. Frances Rivera-Hernández

Co-advisor: Dr. Jacob Adler

Active Galactic Nuclei Research Group, Connecticut College

New London, CT
January 2020 –
August 2021

Description: Extragalactic independent study research, continued over the summer through a Summer Science Research Institute internship, of active galactic nuclei with the goal of better understanding the structure of these objects and investigating the source of their immense energy production.

- Data reduction and calibration of astronomical images
- Differential photometry in the optical and near-infrared regime
- Analyzed light curves of optical and gamma-ray AGN output
- Visually analyzed time delays in different wavebands to monitor activity

Advisor: Dr. Leslie Brown

AWARDS/FELLOWSHIPS

Robin Distinguished Scholar Fellowship, University of Arizona, College of Science

Tucson, AZ
September 2024

Description: Fellowship is awarded for academic excellence, exceptional potential to advance knowledge in the discipline, and ability to broaden perspectives and inquiry based on life experiences.

CONFERENCE PRESENTATIONS¹

* - Poster

** - Talk

Houston, TX
March 2025, *

Singh C., Byrne S., Golombek M. P., Trautman M. R., Dundas C. M., Vona M. A., Williams N. R. (2025). The Boulder Content of Midlatitude Ground Ice on Mars. *56th Annual Lunar and Planetary Science Conference [Abstract #2613]*. The Woodlands, TX.

Pasadena, CA
July 2024, *

Singh C., Byrne S., Golombek M. P., Trautman M. R., Dundas C. M., Williams N. R. (2024). Machine Vision Boulder Detections on Martian Ice Sheets. *Tenth International Conference on Mars [Abstract #3433]*. Pasadena, CA.

Houston, TX
March 2024, *

Singh C., Byrne S., Golombek M. P., Trautman M. R., Dundas C. M., Crocco R., Williams N. R. (2024). The Origin of Boulders on Martian Ice Sheets. *55th Annual Lunar and Planetary Science Conference [Abstract #2755]*. The Woodlands, TX.

¹ All posters, PowerPoint files and/or video recordings are available upon request.

Houston, TX March 2024, *	Singh C. , Biassey-Bogart S., Morris M. M., Williams N. R., Golombek M. P., Do S., Calef III F. J. (2023). Hazard Analysis: Potential Mars Sample Return Landing Sites. <i>55th Annual Lunar and Planetary Science Conference [Abstract #2719]</i> . The Woodlands, TX.
New London, CT October 2022, *	Singh C. , Adler J. B., Rivera-Hernández F. (2022). Characterizing A Potential Remnant of the Hypanis Delta. <i>Summer Science Research Symposium, Connecticut College</i> .
Atlanta, GA July 2022, *, **	Singh C. , Adler J. B., Rivera-Hernández F. (2022). Characterizing A Potential Remnant of the Hypanis Delta. <i>Georgia Tech Broadening Participation in Atmospheric Science, Oceanography, and Geosciences Research Experience for Undergraduates (REU), Georgia Institute of Technology</i>
New London, CT February 2022, *	Singh C. , Brown L. F. (2022). The Optical Monitoring of the Gamma-ray Bright Active Galactic Nucleus 3C345. <i>Summer Science Research Institute Symposium, Connecticut College</i> .
New London, CT July 2021, *	Singh C. , Carbonell W., Modaressi A., Mears W., Brown L. F. (2021). Optical and Gamma-Ray Photometric Monitoring of Active Galactic Nuclei. <i>Summer Science Research Institute Symposium, Connecticut College</i> .
Virtual November 2020, **	Singh C. , Carbonell W., Brown L. F. (2020). The Blazar BL Lacertae: 2018-2020 V, R, and I Band CCD Photometry. <i>Summer Science Research Institute Symposium, Connecticut College</i> .

TEACHING

Geology and Geophysics of the Solar System (PTYS/GEOS/HWRS 411) Guest Lecture, University of Arizona

Tucson, AZ March 2026	• Provided a general overview of glacial and periglacial processes and landforms • Utilized 3D HiRISE anaglyphs to aid students in visualizing glacial and periglacial features on Mars • Led a Q&A session connecting glacial and periglacial concepts on Earth to their Martian analogs
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Academic Resource Center (ARC) Peer Tutor, Connecticut College

New London, CT January 2022 – May 2023	• Ran tutoring sessions for introductory astronomy students • Created a safe, supportive environment while tutoring students • Collaborated with course professor to ensure help sessions met student needs
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FIELD WORK

* - Graduate

** - Undergraduate

* Planetary Geology Field Studies (PTYS 590) – University of Arizona, Fall 2025

- Owens Valley, Eastern Sierra Nevadas:
 - Collected ground penetrating radar (GPR) data measurements of a rock glacier and dunes using pulseEKKO and GSSI equipment and common midpoint and offset techniques

* Planetary Geology Field Studies (PTYS 590) – University of Arizona, Fall 2024

- Channeled Scablands, Eastern Washington state:
 - Obtained drone imagery of boulders and boulder fields to create 3D boulder maps and perform spatial analyses of boulder fields in GIS
 - Collected GPR data measurements of a linguoid bar and gravel dunes using GSSI equipment
 - Recorded boulder compositions and measured boulder heights and diameters to use as ground truth data for remote sensing technique validation

** Geomorphology Labs (GEO 202) – Connecticut College, Fall 2022

- Waterford Beach, Waterford, CT:
 - Used a stadia rod, a tripod, survey tape, and an automatic level to survey coastal area
 - Created a plot of elevation vs. distance to analyze the topography of the area
- Candlewood Hill, Groton, CT:
 - Used a rebound hammer to determine the strength of rock in a ridge-valley system
 - Located and visually analyzed faults in ridge-valley system to interpret formation of area
- Lantern Hill, North Stonington, CT:
 - Used a clinometer to determine the shadow angle and angle of a talus slope
 - Measured rock sizes along talus slope to detect evidence of rock sorting
- Salmon River, Colchester, CT:
 - Located features like meandering channels, terraces, point bars, cut banks, and knickpoints
 - Collected soil cores in various locations and compared samples
- Glacial Park, Ledyard, CT:
 - Examined overall morphology of a local moraine
 - Distinguished between fluvial and glacial processes such as avulsion and sediment sorting

** Plate Tectonics Lab (GEO 201) – Connecticut College, Spring 2022

- Arboretum Ravine, Connecticut College, New London, CT:
 - Worked to detect the presence of a horst and graben system in the area
 - Collected strike and dip measurements of fault zones using a Brunton compass

SERVICE

September 2022 - November 2022	Geoscience Fieldwork Assistant, Connecticut College • Worked in the Connecticut College Arboretum with introductory geology students to complete weekly lab exercises • Led group hikes to field sites • Assisted students in the mastery of basic geology field techniques • Collaborated with course professor to prepare for field exercises
January 2022 - May 2023	Geoscience Assistant, Connecticut College • Organized the Connecticut College rock and mineral collection • Helped photograph and catalog specimens
August 2020 – May 2023	Physics Student Advisory Board, Connecticut College • Elected as chair to serve as a trusted liaison between students and faculty • Planned departmental gatherings and activities for students and faculty

OUTREACH

Tucson, AZ 2026	Tucson Festival of Books (TFoB), Lunar and Planetary Laboratory • Displayed 3D printed terrain models of geologic features on Earth, the Moon, and Mars • Answered attendee questions about the planetary surface features such as Meteor Crater on Earth, Tycho crater on the Moon, and Olympus Mons on Mars • Ran the LPL booth and restocked stickers and mini 3D models
Tucson, AZ 2025, 2026	The Art of Planetary Science (TAPS), Lunar and Planetary Laboratory • Engaged with the public by answering science questions related to artwork • Assisted in setting up displays and advertising the event
Tucson, AZ 2024, 2026	Tucson Gem and Mineral Show (TGMS), Lunar and Planetary Laboratory • Ran glove box demonstration to simulate OSIRIS-REx sample analysis • Led an activity to educate the public about meteor identification

WORKSHOPS

Tempe, AZ May 2024	NASA Planetary Data Training Workshop: GIS, Image Processing, and Data Management, Arizona State University (ASU), School of Earth and Space Exploration (SESE) • Performed radiometric and geometric corrections on raw data products in ISIS • Enhanced ArcGIS Pro and JMARS software and data management proficiency
Tucson, AZ April 2024	NASA Planetary Data Training Workshop: Photogrammetry, University of Arizona (UA), Lunar and Planetary Laboratory (LPL) • Utilized preprocessed images to create DTMs in BAE's SOCET SET • Learned how to generate DTMs using the Ames Stereo Pipeline tool in ISIS